

How We Tested

We assessed the servers on three basic parameters—Design, Features, and Performance.

Design

We took into account the wiring of internal components, proper tying and clamping of wires, expansion bays, tool-free design, ventilation system, etc. We opened up the case and tried installing a new hard drive, removing the existing PCI cards etc., to assess how effortless it is to do these basic things when the server goes down.

Features

Features included processor speeds and type, amount of memory and the speed of memory modules. Hard disk speed and the controller in use, such as SCSI, RAID, etc, make a lot of difference to the I/O speeds.

More emphasis was put on the server management software that should accompany these machines.

Performance

We ran the regular bunch of benchmarking tests that we use on different hardware to gauge performance at subsystem levels.

These subsystem tests included PC Mark 2004—a system suite; SiSoft Sandra; Pov-Ray 3.5—a pure CPU test, the video encoding test and ZD Bench's Content creation 2004.

Being entry level servers, they will perform the role of file servers or Local Intranet servers, or as Database servers. You can use them as mail servers too.

We used PC Mag's NetBench 7.0.3 benchmark program that measures how well a file server handles file I/O requests from 32-bit Windows clients, which pelt the server with requests for network file operations. The final output of NetBench reports throughput measurements.

For gauging the performance of the servers when used as Web servers, we ran PC Mag's WebBench 5.0 benchmarking program. WebBench returns two scores; requests per second and throughput as measured in bytes per second.

For Database performance, we used Oracle 9i as the database application and ran Oratools—Open source tools based on the TPC-C benchmark. We used the ORABM tool which typically answers the question, "How does a server perform under a

CPU/memory intensive Oracle database workload?" The load necessary for benchmarking is generated using the ORABMload script, which creates all the necessary objects such as tables and indices, and loads the data. The final result gives the number of transactions per second the server is capable of.

For the Netbench and WebBench benchmarks we used two clients that pinged the server with requests. To avoid the influence of network characteristics on the results, we did these tests on isolated networks consisting of a Comex DSG-1008, a gigabit 8-port switch and CAT-6 cables for connecting them. We used Windows 2000 Server as the operating system.

The Awards Process

The scores noted from Design, Features, and Performance tests and from the Price Index were given weightage. An overall score out of 100 was calculated. The product that scored the highest in the overall score was adjudged the winner of the Digit Best Buy Gold Award for its category. The runner-up received the Digit Best Buy Silver award.

Zenith's 3 GHz-based server was the fastest processor in this server shootout. Acer's Altos G520 had an Intel 2.8 GHz Xeon processor running at 800 MHz FSB with 1 MB of L2 cache. Hyper-threading was common on all the servers tested.

Memory: Crucial For Performance

The memory subsystem is critical to the performance of a server. Nearly all the servers shipped with 400 MHz memory modules, except for the Acer Altos G520. In the i875P category, PCS was the only one to offer 512 MB of memory, whereas HP and Acer came with just 256 MB. However, none of these servers could take advantage of the dual-channel configuration, since they had only one memory module.

The Servers from Zenith and Wipro were based on the E7210 chipset. They had 1 GB of memory and were configured in dual-channel mode. The Acer Altos G520, also in dual-channel mode, had 512 MB of RAM.

The I/O Subsystem: A chain is only...

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A SCSI system overcomes

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Zenith's Data Server performs well for its price

this problem, but SCSI drives are expensive. They cost nearly twice as much as IDE or SATA drives, and the performance difference is not substantial. But as of now, these drives are the fastest, and until SATA drives mature enough to live up to their potential, SCSI drives will remain the backbone of most servers.

Some of the servers we received had the SATA controllers onboard. Their performance did not match that of SCSI controller cards. Also, an onboard controller will hog CPU time, which doesn't happen with dedicated SCSI cards.

The HP and Wipro servers, as well as the Acer Altos G520, had SCSI hard drives and were hooked up to an Ultra 320 SCSI controller. RAID controllers were also offered on these three models. Wipro bundled three drives, whereas the Altos G520 had two SCSI drives. In both cases the storage space amounted to 80 GB. HP came with just one 40 GB drive.

Zenith offered two SATA hard drives totalling 240 GB. The onboard SATA controller provided the necessary support and RAID features.

Acer's Altos G310 offered two SATA drives, each 80 GB, and these were driven by the onboard SATA and RAID controllers.

The PCS Siirus Compact server offered just one 80 GB drive hooked on to the motherboard's onboard controller.

Of Network Adapters

Nearly all the motherboards had an onboard Gigabit Ethernet controller, freeing up one PCI slot. The Acer Altos G520, the Wipro Netpower and the Zenith Data server all offered both a 10/100 Ethernet and a Gigabit Port. The graphics